Assignment

Question 1: Iris Flower Classification Using MLP

Using the Iris Flower Dataset available in sklearn.datasets.load_iris, implement a Multilayer Perceptron (MLP) classifier from scratch using NumPy, without using predefined MLP algorithms. Use the four input features to classify the flowers into Setosa, Versicolor, and Virginica.

Implement the MLP using three different architectures: 4–6–3, 4–10–3, and 4–20–3, representing 4 input neurons, 6/10/20 hidden neurons, and 3 output neurons, respectively. Demonstrate input feeding, weight and bias initialization, forward propagation, ReLU activation, softmax output, categorical cross-entropy loss, backpropagation, gradient calculation, and weight and bias updates using gradient descent.

Train and test each architecture using 70:30 and 80:20 train–test splits. Report accuracy, precision, recall, F1-score, and confusion matrices. Prepare a performance metrics table for all six experiments, plot the training loss curves, and report the initial and updated weight matrices to demonstrate learning. Discuss the effect of hidden-layer size and training data proportion on classification performance.

Question 2: Breast Cancer Classification Using MLP

Using the Breast Cancer Wisconsin (Diagnostic) Dataset available in sklearn.datasets.load_breast_cancer, implement an MLP classifier from scratch using NumPy. Use the 30 numerical input features to classify tumors as benign or malignant.

Implement and compare three network architectures: 30–8–1, 30–15–1, and 30–25–1, representing 30 input neurons, 8/15/25 hidden neurons, and 1 sigmoid output neuron. Demonstrate input preprocessing, weight and bias initialization, forward propagation, ReLU activation, sigmoid output, binary cross-entropy loss, backpropagation, gradient calculation, and weight and bias updates using gradient descent.

Train and test each architecture using 70:30 and 80:20 train–test splits. Evaluate each model using accuracy, precision, recall, F1-score, and confusion matrices. Prepare a comparative performance metrics table for all six experiments, plot the training loss curves, and report the initial and updated weight matrices to demonstrate the learning process. Discuss the effect of hidden-layer size and training data proportion on classification performance.

Dataset Link:

1. Iris Dataset: https://archive.ics.uci.edu/dataset/53/iris
2. Breast Cancer Dataset: https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic